

Chapter-10

ROTATIONAL MOTION

Moment of Inertia: [I] :- It is the property by virtue of which body opposes the change of state of motion.

(i) For point mass,

$$I = mr^2$$

Where,

$r = \perp^r$ distance from axis of rotation

(ii) For system of masses,

$$I = m_1r_1^2 + m_2r_2^2 + \dots + m_nr_n^2 = \sum_{i=1}^n m_i r_i^2$$

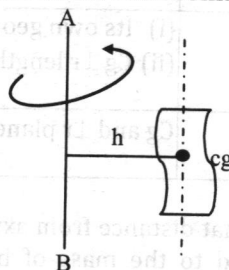
(iii) For body continuous mass distribution,

$$I = \int x^2 dm$$

Where dm = mass of elementary strip

x = distance from axis of rotation to elementary mass.

Parallel Axis Theorem:-



$$I_{AB} = Mh^2 + I_{cg}$$

Perpendicular Axis Theorem:-

The sum of MI of two axes mutually $\perp^r$ in the plane of Lamina gives MI about axis passing through intersection of two axis and $\perp^r$ to the plane of lamina.

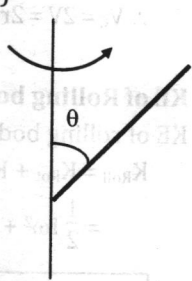
$$\text{i.e. } I_z = I_x + I_y$$

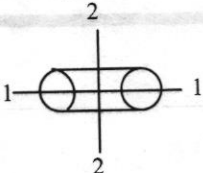
→ Moment of inertia is Tensor.

→ It's unit - kgm^2

Dimension $\rightarrow ML^2$

MI of different bodies:

Bodies	Axis passing through	MI
1. Uniform Rod (M, L)	(i) Cg and $\perp^r$ to length	$\frac{1}{12} ML^2$
	(ii) End and $\perp^r$ to length	$\frac{1}{3} ML^2$
	(iii) 	$\frac{1}{3} ML^2 \sin^2 \theta$
2. Uniform Ring (M, R)	(i) Cg $\perp^r$ to plane	MR^2
	(ii) Diameter	$\frac{1}{2} MR^2$
3. Uniform Disc (M, R)	(i) Cg and $\perp^r$ to plane	$\frac{1}{2} MR^2$
	(ii) Diameter	$\frac{1}{4} MR^2$
4. Solid sphere (M, R)	about diameter	$\frac{2}{5} MR^2$

5. Hollow sphere (M, R)	about diameter	$\frac{2}{3} MR^2$
6. Solid Cylinder	(1) Its own geometrical axis	$\frac{1}{2} MR^2$
	(2) Cg and $\perp$ r length	$M\left(\frac{L^2}{12} + \frac{R^2}{4}\right)$
7. Hollow cylinder	(i) Its own geometrical axis (ii) Cg $\perp$ r length	MR^2 $M\left(\frac{L^2}{12} + \frac{R^2}{2}\right)$
8. Rectangular lamina	Cg and $\perp$ r plane	$\frac{M}{12} (l^2 + b^2)$

Radius of gyration(K): It is that distance from axis of rotation which square, multiplied to the mass of body product gives MI about that axis of rotation.

$$I = mK^2$$

Used only for tangent	$\frac{1}{2}$	Diameter	Ring	Cg $\perp$ r plane	$\frac{1}{1}$	Coefficients of MR^2
	$\frac{1}{4}$	Diameter	Disc	Cg $\perp$ r plane	$\frac{1}{2}$	
					$\frac{2}{5}$	
					$\frac{2}{3}$	
					$\frac{2}{3}$	

$$I_{\text{tangent}} = \left(\frac{N+D}{D}\right) \times MR^2$$

where N = Nominator of Coefficient
D = Dinominator

$$K = \sqrt{\text{coefficient} * 'L' \text{ or } 'R'}$$

eg. (i) For Ring tangent $\perp$ r to plane $\frac{1}{1} \rightarrow$ coefficient

$$N = 1, D = 1 \text{ and } I = 2MR^2$$

(ii) For Ring tangent only $\left(\frac{1}{2}\right) I_{\text{tangent}} = \frac{3}{2} MR^2$

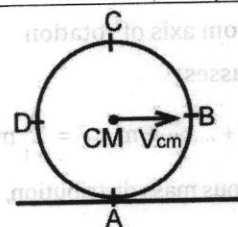
(iii) Radius of gyration for solid sphere about tangent.

$$K = \sqrt{\frac{2+5}{5}} * R = \sqrt{\frac{7}{5}} R$$

(iv) Radius of gyration of uniform rod about are end.

$$K = \sqrt{\frac{1}{3} * L} = \frac{L}{\sqrt{3}}$$

* Plane parallel motion of Rolling body:-



$$V_{\text{cm}} = V \text{ (say)}$$

$$V_A = V - r\omega \quad V_B = \sqrt{V^2 + (r\omega)^2}$$

$$V_C = V + r\omega \quad V_D = \sqrt{V^2 + (r\omega)^2}$$

- If $V > r\omega$, $V_A = +ve$; i.e. it is the case of rolling with slipping and slipping takes place in forward direction.
- If $V < r\omega$, $V_A = -ve$, i.e., it is also the case of rolling with slipping and slipping takes place in backward direction.
- If $V = r\omega$, $V_A = 0$, i.e., it is the case of **rolling without slipping or pure rolling**.

$$\therefore V_C = 2V = 2r\omega, V_B = V_D = \sqrt{2} V = \sqrt{2} r\omega$$

KE of Rolling body:-

KE of rolling body is due to both translation and rotation.

$$K_{\text{Roll}} = K_{\text{Rot}} + K_{\text{Trans}}$$

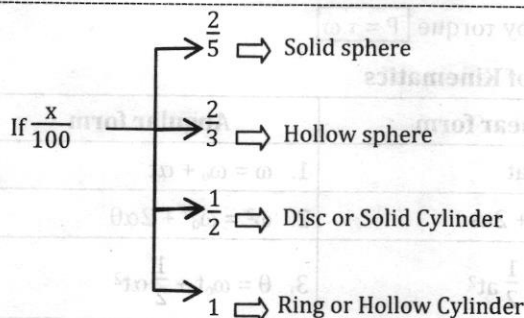
$$= \frac{1}{2} I\omega^2 + \frac{1}{2} mv^2$$

$$K_{\text{Roll}} \text{ or } K_{\text{Total}} = \frac{1}{2} MV^2 * \frac{N+D}{D} = \frac{1}{2} I\omega^2 \frac{N+D}{N}$$

$$\frac{K_{\text{Rot}}}{K_{\text{Roll}} \text{ (or } K_{\text{Total}})} = \frac{N}{N+D} = \frac{K^2}{K^2 + R^2}$$

$$\frac{K_{\text{Trans}}}{K_{\text{Total}} \text{ (or } K_{\text{Roll}})} = \frac{D}{N+D} = \frac{R^2}{R^2 + K^2}$$

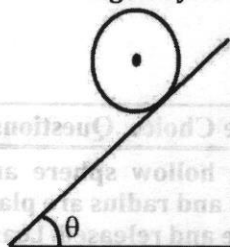
- If KE of a rolling body due to rotational motion is x % of KE due to translatory motion then body must be,



e.g. (i) If a body rolls without slipping, KE due to rotational motion is 40% of KE due to translatory motion then find the body.

$$\frac{40}{100} = \frac{2}{5} = \text{solid sphere}$$

*** Acceleration of Rolling body on inclined plane:**



(1) In case of smooth inclined plane rolling body slides only. i.e., only translation occurs.

$$a = g \sin \theta \quad \theta \Rightarrow \text{Angle of inclination with horizontal}$$

(2) In case of rough inclined plane:-

$$a_{\text{Roll}} = \left(\frac{D}{N+D} \right) * g \sin \theta = \left(\frac{R^2}{K^2+R^2} \right) * g \sin \theta$$

• Velocity when it reaches the bottom from height h .

$$V = \sqrt{2gh * \left(\frac{D}{N+D} \right)}$$

• Velocity when body roll down distance ' l '.

$$v = \sqrt{2g/\sin \theta * \frac{D}{N+D}}$$

• Min^m coefficient of friction required for rolling

$$\mu_{\min} = \tan \theta * \left(\frac{N}{N+D} \right) = \left(\frac{K^2}{K^2+R^2} \right) * \tan \theta$$

Bodies	Acceleration	Min ^m coefficient of friction
Ring $\left(\frac{1}{1} \right)$	$g \sin \theta * \frac{1}{2}$	$\tan \theta * \frac{1}{2}$
Disc $\left(\frac{1}{2} \right)$	$g \sin \theta * \frac{2}{3}$	$\tan \theta * \frac{1}{3}$

Solid sphere $\left(\frac{2}{5} \right)$	$g \sin \theta * \frac{5}{7}$	$\tan \theta * \frac{2}{7}$
Hollow sphere $\left(\frac{2}{3} \right)$	$g \sin \theta * \frac{3}{5}$	$\tan \theta * \frac{2}{5}$

$$t \propto \frac{1}{\sqrt{a}} ; v \propto \sqrt{a}$$

$$a_{\text{solid sphere}} > a_{\text{Disc}} > a_{\text{H. sphere}} > a_{\text{Ring}}$$

$$t_{\text{S. sphere}} < t_{\text{Disc}} < t_{\text{H. sphere}} < t_{\text{Ring}}$$

• If a body rolls to up the hill with velocity v then maximum height attained.

$$h = \frac{v^2}{2g} * \left(\frac{N+D}{D} \right)$$

• Acceleration of mass tied to a string wound over a cylinder & mounted on a horizontal axis is

$$a = g * \left(\frac{D}{N+D} \right)$$

• Tension in the string $T = mg * \left(\frac{D}{N+D} \right)$

• If ball of radius ' R ' is rotating with angular velocity ' ω_0 ' is placed on rough horizontal surface having the coefficient of friction μ , then ball starts pure rolling (i.e., rolling without slipping).

(i) After time, $t = \frac{2 \omega_0 R}{7 \mu g} = \frac{2 \omega_0 R}{7 \mu g}$

(ii) The distance after which

$$s = \frac{2 \omega_0^2 R^2}{49 \mu g}$$

• The temperature of body of MI ' I ' having coefficient of linear expansion ' α ' is increased by ' Δt ' then change in moment of inertia is given by

$$\Delta I = I_t - I = 2I \alpha \Delta t$$

• If two solid sphere are of same material having their radius R_1 and R_2 respectively then this ratio of MI.

$$\frac{I_1}{I_2} = \left(\frac{R_1}{R_2} \right)^5 ; I = \frac{8}{15} \pi \rho R^5$$

• Moment of inertia depends upon mass distribution of body about axis of rotation and mass of the body.

• A bicycle is fitted with spokes to increase in MI of wheel that's cause balance of cyclist is maintained.

• MI of raw egg > MI of boiled egg.

- If a rod of length ' l ' is hinged to lower end and allowed to fall then free end strikes the floor at velocity

$$v = \sqrt{3gl}$$

Angular momentum ($\vec{L}$): $\rightarrow$

$$|\vec{L}| = |\vec{r} \times \vec{p}| = rps \sin \theta$$

$L = p \times 1r$ distance = Moment of linear momentum.

$$L = I\omega \Rightarrow \vec{L} = I\vec{\omega}$$

KE due to rotation. $K = \frac{1}{2}I\omega^2 = \frac{1^2}{2I} = \pi Lf$

$$L \propto \frac{K}{f}$$

Torque ($\vec{\tau}$): $\vec{\tau} = \vec{r} \times \vec{F}$

$\tau = F \times 1r$ distance = Moment of force

$$\tau = I\alpha = \frac{dL}{dt}$$

If angular momentum linearly varying then

$$\tau = \frac{L_f - L_i}{t}$$

- If external torque acting on system is zero then angular momentum is conserved (i.e., constant)

$$L = \text{constant} \Rightarrow I\omega = \text{constant}$$

$$I_1\omega_1 = I_2\omega_2 = \dots I_n\omega_n$$

$$\omega \propto \frac{1}{I} \text{ and } T \propto I \text{ where } T = \text{time period}$$

i.e. if $I \uparrow$ ses, $T \uparrow$ ses & $\omega \downarrow$ ses.

- If polar ice melts, then length of day increases.
- If the radius of earth becomes R/n without changing its mass then time period of rotation becomes

$$T' = \frac{24}{n^2} \text{ hr} \quad I \propto R^2 \propto T$$

$$\text{For } R/2 : n = 2, T' = 6 \text{ hr}$$

$$\text{For } R/3 : n = 3, T' = 8/3 \text{ hr}$$

- A swimmer while jumping into water from height easily forms a loop in air if he pulls his arms and leg in.

Couple:

Two equal and opposite forces separated at a distance.

- couple produces purely rotation.

- work done by torque, $\omega = \int_{\theta_1}^{\theta_2} \tau d\theta$

power by torque $P = \tau \cdot \omega$

Equation of Kinematics

Linear form	Angular form
1. $v = u + at$	1. $\omega = \omega_0 + \alpha t$
2. $v^2 = u^2 + 2as$	2. $\omega^2 = \omega_0^2 + 2\alpha\theta$
3. $s = ut + \frac{1}{2}at^2$	3. $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$
4. $s = \frac{(u+v)}{2}at$	4. $\theta = \frac{(\omega_0 + \omega)}{2} * t$

Tangential acceleration (a_t) = $r\alpha$

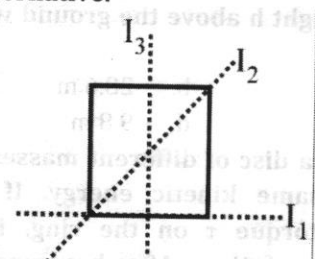
$$\text{No. of revolution} = \frac{\theta(c)}{2\pi} \text{ or } \frac{\theta^{(0)}}{360^\circ}$$

$$\langle \omega \rangle = \frac{\Delta\theta}{\Delta t}, \quad \omega = \frac{d\theta}{dt}$$

$$\langle \alpha \rangle = \frac{\Delta\omega}{\Delta t}; \quad \alpha = \frac{d\omega}{dt}$$

Multiple Choice Questions

- A solid sphere, a hollow sphere and a disc, all having same mass and radius are placed at the top of a smooth incline and released. Least time will be taken in reaching the bottom by:
 - the solid sphere
 - the hollow sphere
 - the disc
 - all will take same time
- If a spherical ball rolls on a table without slipping, the fraction of its total energy associated with rotation is:
 - $3/5$
 - $2/7$
 - $2/5$
 - $3/7$
- The speed of a homogeneous solid sphere after rolling down an inclined plane of vertical height h , from rest without sliding is:
 - $\sqrt{gh}$
 - $\sqrt{\left(\frac{g}{5}\right)gh}$
 - $\sqrt{\left(\frac{4}{3}\right)gh}$
 - $\sqrt{\left(\frac{10}{7}\right)gh}$
- Moment of inertia of a rod of mass m and length l about its one end is I . If one fourth of its length is cut away then moment of inertia of the remaining rod about its one end will be:
 - $\frac{3}{4}I$
 - $\frac{9}{16}I$
 - $\frac{27}{64}I$
 - $\frac{1}{16}I$
- When a body is projected at an angle horizontal in the uniform gravitational 'earth, the angular momentum of the out

- the point of projection, as it proceeds along its path:
- remains constant
 - increases
 - decreases
 - initially decreases and after its highest point increases
6. A force F acts tangentially at the highest point of the sphere of mass M kept on a rough horizontal plane. If the sphere rolls without slipping, the acceleration of the centre of sphere is:
- $\frac{10F}{7M}$
 - $\frac{7F}{10M}$
 - $\frac{10M}{7F}$
 - $\frac{10}{7}MF$
7. A rigid body rotates about a fixed axis with variable angular velocity equal to $\alpha - \beta t$, at time t where α, β are constants. The angle through which it rotates before it stops:
- $\frac{\alpha^2}{2\beta}$
 - $\frac{\alpha^2 - \beta^2}{2\alpha}$
 - $\frac{\alpha^2 - \beta^2}{2\beta}$
 - $\frac{(\alpha - \beta)\alpha}{2}$
8. A rod of length ' L ' whose lower end is fixed along the horizontal plane starts to topple from the vertical position. The velocity of the upper end of the rod when it hits the ground is:
- $\sqrt{gh}$
 - $\sqrt{3gL}$
 - $\sqrt{5gL}$
 - $3\sqrt{gL}$
9. Three thin rods each of length ' L ' and mass M are placed along X, Y, and Z axis such that one end of each rod is at origin. The moment of inertia of this system about z-axis is:
- $\frac{2}{3}ML^2$
 - $\frac{4ML^2}{3}$
 - $\frac{5ML^2}{3}$
 - $\frac{ML^2}{3}$
10. A constant torque acting on a uniform circular wheel changes its angular momentum from A_0 to $4A_0$ in 4 second. The magnitude of the torque is:
- $\frac{3A_0}{4}$
 - $\frac{A_0}{4}$
 - $4A_0$
 - $12A_0$
11. A force of $(3\hat{i} + 4\hat{k})$ N is acted upon a fly wheel tangentially. The radius vector of the flywheel has $(2\hat{j} - 6\hat{k})$ m. The torque acting on the fly wheel is:
- 25 N-m
 - 20.6 N-m
 - $\sqrt{42}$ N-m
 - none of these
12. A solid sphere and a disc of same mass and radius starts rolling down a rough inclined plane, from the same height the ratio of the time taken in the two cases is:
- 15 : 14
 - $\sqrt{15} : \sqrt{14}$
 - 14 : 15
 - $\sqrt{14} : \sqrt{15}$
13. A particle is moving in a circular orbit with constant speed select wrong alternate:
- Its momentum is conserved
 - Its angular momentum is conserved
 - It is moving with variable velocity
 - It is moving with variable acceleration
14. A square lamina is as shown in figure. The moment of inertia of the frame about the three axes shown in figure are I_1, I_2 and I_3 respectively. Select the correct alternative.
- 
- $I_2 = I_3 > I_1$
 - $I_1 > I_2 > I_3$
 - $I_2 = I_3 < I_1$
 - $I_1 < I_2 < I_3$
15. Three identical rods, each of length L , are joined to form a rigid equilateral triangle. Its radius of gyration about an axis passing through a corner and perpendicular to the plane of the triangle is:
- $\frac{L}{2}$
 - $\sqrt{\frac{3}{2}}L$
 - $\frac{L}{\sqrt{2}}$
 - $\frac{L}{\sqrt{3}}$
16. A solid sphere of mass 2 kg rolls up a 30° incline with an initial speed of 10 ms^{-1} . The maximum height reached by the sphere is ($g = 10 \text{ ms}^{-2}$)
- 3.5 m
 - 7.0 m
 - 10.5 m
 - 14.0 m
17. A ring and a disc having the same mass, roll without slipping with the same linear velocity. If the kinetic energy of the ring is 8J, that of the disc must be:
- 2J
 - 4J
 - 6J
 - 16J
18. A flywheel having a radius of gyration of 2m and mass 10 kg rotates at an angular speed of rad/s about an axis perpendicular to it through its centre. The kinetic energy of rotation is:
- 500 J
 - 2000 J
 - 1000 J
 - 250 J
19. If a disc of mass m and radius r is reshaped into a ring of radius $2r$, the mass remaining the same, the radius of gyration goes up by a factor of:
- $\sqrt{2}$
 - 2
 - $2\sqrt{2}$
 - 4
20. A uniform cylinder has radius R and length L . If the moment of inertia of this cylinder about an axis passing through its centre and normal to its

circular face is equal to the moment of inertia of the same cylinder about an axis passing through its centre and normal to its length; then:

- a. $L = R$ b. $L = \sqrt{3} R$
c. $L = \frac{R}{\sqrt{3}}$ d. $L = 0$

21. A uniform solid sphere rolls on a horizontal surface at 20 ms^{-1} . It then rolls up an incline having an angle of inclination at 30° with the horizontal. If the friction losses are negligible, the value of height h above the ground where the ball stops is:

- a. 14.3 m b. 28.6 m
c. 57.2 m d. 9.8 m

22. A ring and a disc of different masses are rotating with the same kinetic energy. If we apply a retarding torque τ on the ring, it stops after making n revolutions. After how many revolutions will the disc stop if the retarding torque on it is also τ ?

- a. $n/2$ b. n
c. $2n$ d. $4n$

23. A sphere of mass M rolls without slipping on an inclined plane of inclination θ . What should be the minimum coefficient of friction, so that the sphere rolls down without slipping?

- a. $\frac{2}{5} \tan \theta$ b. $\frac{2}{7} \tan \theta$
c. $\frac{5}{7} \tan \theta$ d. $\tan \theta$

24. A body of mass M and radius R is rolling horizontally without slipping with speed v . It then rolls up a hill to a maximum height h . If $h = 5v^2/6g$, what is the moment of inertia of the body?

- a. $\frac{1}{2} MR^2$ b. $\frac{2}{3} MR^2$
c. $\frac{3}{4} MR^2$ d. $\frac{2}{5} MR^2$

25. A couple produces:

- a. no motion
b. linear and rotational motion
c. purely rotational motion
d. purely linear motion

26. P is the point of contact of a wheel and the ground. the radius of wheel is 1m. The wheel rolls on the ground without slipping. The displacement of point P when wheel completes half rotation is:

- a. 2m b. $\sqrt{\pi^2 + 4}$ m
c. π m d. $\sqrt{\pi^2 + 2}$ m

27. A body rolls down an inclined plane. If its kinetic energy of rotational motion is 40% of its kinetic energy of translation motion, then the body is a:

- a. solid sphere b. spherical shell
c. cylinder d. ring

28. A particle performs uniform circular motion with an angular momentum L . If the frequency of the particle motion is double and its kinetic energy is halved the angular momentum becomes:

- a. $\frac{L}{4}$ b. $\frac{L}{2}$
c. $2L$ d. $4L$

29. A circular disc is rotating with angular velocity ω . If a man standing at the edge of the disc walks towards its centre, then the angular velocity of the disc:

- a. is not changed b. be halved
c. decreases d. increases

30. When a disc having velocity of its centre of mass v_{cm} is rolling, then velocity of the highest point on the disc will be:

- a. $2v_{cm}$ and at point of contact is v_{cm}
b. $3v_{cm}$ and at point of contact is v_{cm}
c. $2v_{cm}$ and at point of contact is zero
d. v_{cm} and at point of contact is v_{cm}

31. The moment of inertia of a body about a given axis is 1.2 kg-m^2 . To produce a rotational kinetic energy of 1500 J an angular acceleration of 25 rad/sec^2 must be applied for:

- a. 8.5 sec b. 5 sec
c. 2 sec d. 1 sec

32. The angular speed of a body changes from ω_1 to ω_2 without applying a torque but due to change in its moment of inertia. The ratio of radii of gyration in the two cases is:

- a. $\omega_2 : \omega_1$ b. $\omega_1 : \omega_2$
c. $\sqrt{\omega_1} : \sqrt{\omega_2}$ d. $\sqrt{\omega_2} : \sqrt{\omega_1}$

33. A coin, placed on a rotating turn-table stops, when it is placed at a distance of 9 cm from its centre. If the angular velocity of the turn table is tripled, it will just stop at a distance r from centre. The value of r is:

- a. 1 cm b. 3 cm
c. 9 cm d. 27 cm

34. A wheel is rotating at the rate of 33 rev/min. If it comes to stop in 20 sec. Then the angular retardation will be:

- a. zero b. $11\pi \text{ rad/s}^2$
c. $\frac{\pi}{200} \text{ rad/s}^2$ d. $\frac{11\pi}{200} \text{ rad/s}^2$

35. The angular velocity of the wheel increases from 1200 to 4500 rev per min in 10 seconds. Then the number of revolutions made during this time is:

- a. 575 b. 675
c. 875 d. 475

Answers:

1. d	2. b	3. d	4. c	5. b
6. a	7. a	8. b	9. a	10. a
11. b	12. d	13. a	14. c	15. b
16. b	17. c	18. a	19. c	20. b
21. b	22. b	23. b	24. b	25. c
26. b	27. a	28. a	29. b	30. c
31. a	32. d	33. a	34. d	35. d

Solution for Selected Problems

2. $\frac{K_{\text{Rot}}}{K_{\text{total}}} = \frac{N}{N+D}$ or $\frac{K^2}{k^2 + R^2}$

For solid sphere: $I = \frac{2}{5} MR^2$

Coefficient = $\frac{2}{5}$

$\frac{K_{\text{Rot}}}{K_{\text{total}}} = \frac{2}{2+5} = \frac{2}{7}$

3. Velocity when rolled down from height 'h'

$V = \sqrt{2gh \left(\frac{D}{N+D} \right)} = \sqrt{2gh \left(\frac{5}{2+5} \right)} = \sqrt{\frac{10gh}{7}}$

4. $I = \frac{1}{3} ML^2$

$I' = \frac{1}{3} M'L'^2 = \frac{1}{3} \left(\frac{3M}{4} \right) \times \left(\frac{4}{3} \right)^2 = \frac{1}{3} ML^2 \times \frac{27}{64}$

$I' = \frac{27}{64} I$

6.

$\Sigma F = ma$

$F + f = ma$ (i)

$\Sigma \tau = I\alpha$

$\Rightarrow (F - f)R = \frac{2}{5} MR^2 \times \alpha$

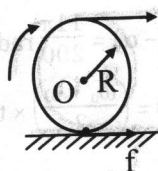
$F - f = \frac{2}{5} Ma$ (ii)

$[a = R\alpha]$

Adding (i) and (ii) we get,

$a = \frac{10F}{7M}$

7. $\omega = \alpha - \beta t$



The time after which body stops rotation, $t = \frac{\alpha}{\beta}$

Again, $\omega = \alpha - \beta t$

$\frac{d\theta}{dt} = \alpha - \beta t$

$\Rightarrow \int_0^\theta d\theta = \int_0^{\alpha/\beta} (\alpha - \beta t) dt$

$\therefore \theta = \left[\alpha t - \frac{\beta t^2}{2} \right]_0^{\alpha/\beta} = \frac{\alpha^2}{2\beta}$

8. loss in PE = Gain in KE

$\Rightarrow \frac{1}{2} mgL = \frac{1}{2} I\omega^2$

$\therefore \frac{1}{2} mgL = \frac{1}{2} \times \frac{1}{3} ML^2 \times \frac{V^2}{L^2}$

$\therefore V = \sqrt{3gL}$

9. The rod which is on Z-axis, its MI about Z-axis is zero.

$I_z = \frac{1}{3} ML^2 + \frac{1}{3} ML^2 = \frac{2}{3} ML^2$

10. $\vec{\tau} = \vec{r} \times \vec{F}$

$\vec{\tau} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 2 & -6 \\ 3 & 0 & 4 \end{vmatrix}$

$\therefore \vec{\tau} = 8\vec{i} - \vec{j} \times (-18) + \vec{k} \times (-6) = 8\vec{i} + 18\vec{j} - 6\vec{k}$

$t = \sqrt{8^2 + 18^2 + (-6)^2} = 20.6 \text{ Nm}$

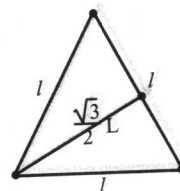
12. $t \propto \frac{1}{\sqrt{a}}$

For Disc: $a = g \sin \theta \times \frac{2}{3}$

For solid sphere: $a = g \sin \theta \times \frac{5}{7}$

$\frac{t_{\text{solid sphere}}}{t_{\text{Disc}}} = \sqrt{\frac{g \sin \theta \times 2/3}{g \sin \theta \times 5/7}} = \sqrt{\frac{14}{15}}$

15.



Moment of Inertia passing through one corner and perpendicular to plane.

$$I = \frac{1}{3} ML^2 + \frac{1}{3} ML^2 + M \left(\frac{\sqrt{3}L}{2} \right)^2 + \frac{1}{12} ML^2$$

$$= \left(\frac{4+4+9+1}{12} \right) ML^2 = \frac{18}{12} ML^2 = \frac{3}{2} ML^2$$

$$I = MK^2 = \frac{3}{2} ML^2$$

$$\therefore K = \frac{\sqrt{3}}{2} L$$

$$16. h = \frac{v^2}{2g} \times \left(\frac{N+D}{D} \right) = \frac{10^2}{2 \times 10} \times \left(\frac{2+5}{5} \right) = 7\text{m}$$

17. kinetic energy of ring:

$$k = \frac{1}{2} mv^2 \times \left(\frac{1+1}{1} \right) = mv^2$$

$$\text{kinetic energy of Disc; } k' = \frac{1}{2} mv^2 \times \frac{1}{2} = \frac{1}{2} \times \frac{3}{2} \times 8 = 6\text{ J}$$

$$18. k = \frac{1}{2} I \omega^2 = \frac{1}{2} MK^2 \times \omega^2$$

$$= \frac{1}{2} \times 10 \times 2^2 \times 5^2 = 500\text{J}$$

$$19. \text{Radius gyration for Disc: } k = \frac{1}{\sqrt{2}} R$$

Again, Radius of gyration for ring of radius $2R$, $k' = 2R$

$$k' = 2 \times \sqrt{2} k = 2\sqrt{2} k$$

$$\therefore \frac{k'}{k} = 2\sqrt{2}$$

$$20. \frac{1}{2} MR^2 = M \left[\frac{R^2}{4} + \frac{L^2}{12} \right]$$

$$\therefore L = \sqrt{3} R$$

$$21. h = \frac{v^2}{2g} \times \frac{7}{5} = \frac{2.2}{(2 \times 9.8)} \times \frac{7}{5} = 28.6\text{m}$$

$$22. \tau \times \theta = \frac{1}{2} I \omega^2$$

$$\text{And no. of revolution, } n = \frac{\theta}{2\pi}$$

$$\therefore \tau \times 2\pi n = k$$

$$n = \frac{k}{2\pi\tau}$$

$$n \propto k$$

Hence no. of revolutions for Disc = n

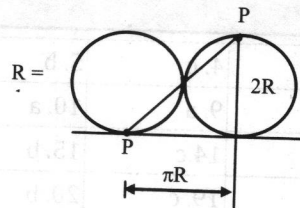
$$24. h = \frac{5v^2}{69} = \frac{v^2}{2g} \times \frac{5}{3}$$

$$\text{Max}^m \text{ height: } h = \frac{v^2}{2g} \times \left(\frac{N+D}{D} \right)$$

$$\text{comparing; } \frac{N}{D} = \frac{2}{3}$$

$$MI = \frac{2}{3} MR^2$$

26.



$$\text{Displacement of point P} = \sqrt{(\pi R)^2 + (2R)^2} = \sqrt{\pi^2 + 4}$$

$$28. L \propto \frac{k}{f}$$

$$L' \propto \frac{k'}{f'} \propto \frac{(k/2)}{2f}$$

$$\therefore L' = \frac{L}{4}$$

$$32. \tau = 0$$

Angular momentum (L) = constant

$$L = I\omega = MK^2 \times \omega$$

$$\therefore k \propto \frac{1}{\sqrt{\omega}}$$

$$\frac{k_1}{k_2} = \frac{\sqrt{\omega_2}}{\sqrt{\omega_1}}$$

$$33. r \propto \frac{1}{\omega^2} \left[r = \frac{\mu g}{\omega^2} \right]$$

$$\therefore \frac{r_2}{r_1} = \left(\frac{\omega_1}{\omega_2} \right)^2$$

$$\therefore r_2 = 1\text{cm}$$

$$34. \omega = 33\text{rev/min}$$

$$= 33 \times \frac{2\pi}{60} \text{ rad/sec}$$

$$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{0 - (33 \times 2\pi/60)}{20}$$

$$\therefore (-\alpha) = \frac{11\pi}{200} \text{ rad/s}^2$$

$$35. \theta = \left(\frac{\omega_0 + \omega}{2} \right) \times t = \frac{(5700 \times 2\pi/60)}{2} \times 10 = 950\pi$$

$$\text{no. of revolution (n)} = \frac{\theta}{2\pi} = 475$$

